Annex A (normative): Measurement channels

A.1 General

The throughput values defined in the measurement channels specified in Annex A, are calculated and are valid per datastream (codeword). For multi-stream (more than one codeword) transmissions, the throughput referenced in the minimum requirements is the sum of throughputs of all datastreams (codewords).
The UE category entry in the definition of the reference measurement channel in Annex A is only informative and reveals the UE categories, which can support the corresponding measurement channel. Whether the measurement channel is used for testing a certain UE category or not is specified in the individual minimum requirements.

A.2 UL reference measurement channels for E-UTRA TDD Config 2

A.2.1 General

The measurement channels in the following clauses are defined to derive the requirements in clause 6 (Transmitter Characteristics) and clause 7 (Receiver Characteristics). The measurement channels represent example configurations of physical channels for different data rates.

A.2.2 Reference measurement channels for E-UTRA

A.2.2.1 Full RB allocation

A.2.2.1.1 QPSK

Table A.2.2.1.1-1: Reference Channels for QPSK with full RB allocation

Parameter	Unit	Value					
Channel bandwidth	MHz	1.4	3	5	10	15	20
Allocated resource blocks		6	15	25	50	75	100
Uplink-Downlink Configuration (NOTE 2)		2	2	2	2	2	2
Special subframe configuration (NOTE 3)		7	7	7	7	7	7
DFT-OFDM Symbols per Sub-Frame		12	12	12	12	12	12
Modulation		QPSK	QPSK	QPSK	QPSK	QPSK	QPSK
Target Coding rate		1/3	1/3	1/3	1/3	1/5	1/6
Payload size							
For Sub-Frame 2,7	Bits	600	1544	2216	5160	4392	4584
Transport block CRC	Bits	24	24	24	24	24	24
Number of code blocks per Sub-Frame (NOTE 1)							
For Sub-Frame 2,7		1	1	1	1	1	1
Total number of bits per Sub-Frame							
For Sub-Frame 2,7	Bits	1728	4320	7200	14400	21600	28800
Total symbols per Sub-Frame							

For Sub-Frame 2,7		864	2160	3600	7200	10800	14400
UE Category		≥ 1	≥ 1	≥ 1	≥ 1	≥ 1	≥ 1
NOTE 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) NOTE 2: As per Table 4.2-2 in TS 36.211 [7] NOTE 3: As per Table 4.2-1 in TS 36.211 [7]							

A.2.2.1.2 16-QAM

Table A.2.2.1.2-1: Reference Channels for 16-QAM with full RB allocation

Parameter	Unit	Value					
Channel bandwidth	MHz	1.4	3	5	10	15	20
Allocated resource blocks		6	15	25	50	75	100
Uplink-Downlink Configuration (NOTE 2)		2	2	2	2	2	2
Special subframe configuration (NOTE 3)		7	7	7	7	7	7
DFT-OFDM Symbols per Sub-Frame		12	12	12	12	12	12
Modulation		16QAM	16QAM	16QAM	16QAM	16QAM	16QAM
Target Coding rate		3/4	1/2	1/3	3/4	1/2	1/3
Payload size							
For Sub-Frame 2,7	Bits	2600	4264	4968	21384	21384	19848
Transport block CRC	Bits	24	24	24	24	24	24
Number of code blocks per Sub-Frame (NOTE 1)							
For Sub-Frame 2,7		1	1	1	4	4	4
Total number of bits per Sub-Frame							
For Sub-Frame 2,7	Bits	3456	8640	14400	28800	43200	57600
Total symbols per Sub-Frame							
For Sub-Frame 2,7		864	2160	3600	7200	10800	14400
UE Category		≥ 1	≥ 1	≥ 1	≥ 2	≥ 2	≥ 2
NOTE 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) NOTE 2: As per Table 4.2-2 in TS 36.211 [7] NOTE 3: As per Table 4.2-1 in TS 36.211 [7]							

A.2.2.1.3 64-QAM

Table A.2.2.1.3-1: Reference Channels for 64-QAM with full RB allocation

Parameter	Unit	Value					
Channel bandwidth	MHz	1.4	3	5	10	15	20
Allocated resource blocks		6	15	25	50	75	100
Uplink-Downlink Configuration (NOTE 2)		2	2	2	2	2	2
Special subframe configuration (NOTE 3)		7	7	7	7	7	7
DFT-OFDM Symbols per Sub-Frame		12	12	12	12	12	12
Modulation		64QAM	64QAM	64QAM	64QAM	64QAM	64QAM
Target Coding rate		3/4	3/4	3/4	3/4	3/4	3/4
Payload size							
For Sub-Frame 2,7	Bits	3752	9528	15840	31704	46888	63776
Transport block CRC	Bits	24	24	24	24	24	24
Number of code blocks per Sub-Frame (NOTE 1)							
For Sub-Frame 2,7		1	2	3	6	8	11
Total number of bits per Sub-Frame							
For Sub-Frame 2,7	Bits	5184	12960	21600	43200	64800	86400
Total symbols per Sub-Frame							
For Sub-Frame 2,7		864	2160	3600	7200	10800	14400
UE Category (NOTE 4)		5, 8	5, 8	5, 8	5, 8	5, 8	5, 8

UE UL Cateogry (NOTE 4)		5, 8, 13, 14	5, 8, 13, 14	5, 8, 13, 14	5, 8, 13, 14	5, 8, 13, 14	5, 8, 13, 14
NOTE 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) NOTE 2: As per Table 4.2-2 in TS 36.211 [7] NOTE 3: As per Table 4.2-1 in TS 36.211 [7] NOTE 4: If UE does not report UE UL category, then the applicability of reference channel is determined by UE category. If UE reports UE UL category, then the applicability of reference channel is determined by UE UL category.							

A.2.2.1.4 256 QAM

Table A.2.2.1.4-1: Reference Channels for 256 QAM with full RB allocation

Parameter	Unit	Value					
Channel bandwidth	MHz	1.4	3	5	10	15	20
Allocated resource blocks		6	15	25	50	75	100
Uplink-Downlink Configuration (NOTE 2)		2	2	2	2	2	2
Special subframe configuration (NOTE 3)		7	7	7	7	7	7
DFT-OFDM Symbols per Sub-Frame		12	12	12	12	12	12
Modulation		256QAM	256QAM	256QAM	256QAM	256QAM	256QAM
Target Coding rate		3/4	3/4	3/4	3/4	3/4	3/4
Payload size							
For Sub-Frame 2,7	Bits	5160	12960	21384	42368	63776	84760
Transport block CRC	Bits	24	24	24	24	24	24
Number of code blocks per Sub-Frame (NOTE 1)							
For Sub-Frame 2,7		1	3	4	8	11	15
Total number of bits per Sub-Frame							
For Sub-Frame 2,7	Bits	6912	17280	28800	57600	86400	115200
Total symbols per Sub-Frame							
For Sub-Frame 2,7		864	2160	3600	7200	10800	14400
UE UL Cateogry		≥ 15	≥ 15	≥ 15	≥ 15	≥ 15	≥ 15
NOTE 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) NOTE 2: As per Table 4.2-2 in TS 36.211 [7] NOTE 3: As per Table 4.2-1 in TS 36.211 [7]							

A.2.2.2 Partial RB allocation

A.2.2.2.1 QPSK

Table A.2.2.2.1-1: Reference Channels for QPSK with partial RB allocation

Parameter	Ch BW	Allocated RBs	UL-DL Configuration (NOTE 2)	Special subframe configuration (NOTE 3)	DFT-OFDM Symbols per Sub-Frame	Mod'n	Target Coding rate	Payload size for Sub-Frame 2, 7	Transport block CRC	Number of code blocks per Sub-Frame (NOTE 1)	Total number of bits per Sub-Frame for Sub-Frame 2, 7	Total symbols per Sub-Frame for Sub-Frame 2, 7	UE Category
Unit	MHz							Bits	Bits		Bits		
	1.4 - 20	1	2	7	12	QPSK	1/3	72	24	1	288	144	≥1
	1.4 - 20	2	2	7	12	QPSK	1/3	176	24	1	576	288	≥1
	1.4 - 20	3	2	7	12	QPSK	1/3	256	24	1	864	432	≥1
	1.4 - 20	4	2	7	12	QPSK	1/3	392	24	1	1152	576	≥1
	1.4 - 20	5	2	7	12	QPSK	1/3	424	24	1	1440	720	≥1
	3-20	6	2	7	12	QPSK	1/3	600	24	1	1728	864	≥1
	3-20	8	2	7	12	QPSK	1/3	808	24	1	2304	1152	≥1

	3-20	9	2	7	12	QPSK	1/3	776	24	1	2592	1296	≥1
	3-20	10	2	7	12	QPSK	1/3	872	24	1	2880	1440	≥1
	3-20	12	2	7	12	QPSK	1/3	1224	24	1	3456	1728	≥1
	5-20	15	2	7	12	QPSK	1/3	1320	24	1	4320	2160	≥1
	5-20	16	2	7	12	QPSK	1/3	1384	24	1	4608	2304	≥1
	5-20	18	2	7	12	QPSK	1/3	1864	24	1	5184	2592	≥1
	5-20	20	2	7	12	QPSK	1/3	1736	24	1	5760	2880	≥1
	5-20	24	2	7	12	QPSK	1/3	2472	24	1	6912	3456	≥1
	10-20	25	2	7	12	QPSK	1/3	2216	24	1	7200	3600	≥1
	10-20	27	2	7	12	QPSK	1/3	2792	24	1	7776	3888	≥1
	10-20	30	2	7	12	QPSK	1/3	2664	24	1	8640	4320	≥1
	10-20	32	2	7	12	QPSK	1/3	2792	24	1	9216	4608	≥1
	10-20	36	2	7	12	QPSK	1/3	3752	24	1	10368	5184	≥1
	10-20	40	2	7	12	QPSK	1/3	4136	24	1	11520	5760	≥1
	10-20	45	2	7	12	QPSK	1/3	4008	24	1	12960	6480	≥1
	10-20	48	2	7	12	QPSK	1/3	4264	24	1	13824	6912	≥1
	15 - 20	50	2	7	12	QPSK	1/3	5160	24	1	14400	7200	≥1
	15 - 20	54	2	7	12	QPSK	1/3	4776	24	1	15552	7776	≥1
	15 - 20	60	2	7	12	QPSK	1/4	4264	24	1	17280	8640	≥1
	15 - 20	64	2	7	12	QPSK	1/4	4584	24	1	18432	9216	≥1
	15 - 20	72	2	7	12	QPSK	1/4	5160	24	1	20736	10368	≥1
	20	75	2	7	12	QPSK	1/5	4392	24	1	21600	10800	≥1
	20	80	2	7	12	QPSK	1/5	4776	24	1	23040	11520	≥1
	20	81	2	7	12	QPSK	1/5	4776	24	1	23328	11664	≥1
	20	90	2	7	12	QPSK	1/6	4008	24	1	25920	12960	≥1
	20	96	2	7	12	QPSK	1/6	4264	24	1	27648	13824	≥1
NOTE 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) NOTE 2: As per Table 4.2-2 in TS 36.211 [7] NOTE 3: As per Table 4.2-1 in TS 36.211 [7]													

A.2.2.2.2 16-QAM

Table A.2.2.2.2-1: Reference Channels for 16QAM with partial RB allocation

Parameter	Ch BW	Allocated RBs	UL-DL Configuration (NOTE 2)	Special subframe configuration (NOTE 3)	DFT-OFDM Symbols per Sub-Frame	Mod'n	Target Coding rate	Payload size for Sub-Frame 2, 7	Transport block CRC	Number of code blocks per Sub-Frame (NOTE 1)	Total number of bits per Sub-Frame for Sub-Frame 2, 7	Total symbols per Sub-Frame for Sub-Frame 2, 7	UE Category
Unit	MHz							Bits	Bits		Bits		
	1.4 - 20	1	2	7	12	16QAM	3/4	408	24	1	576	144	≥1
	1.4 - 20	2	2	7	12	16QAM	3/4	840	24	1	1152	288	≥1
	1.4 - 20	3	2	7	12	16QAM	3/4	1288	24	1	1728	432	≥1
	1.4 - 20	4	2	7	12	16QAM	3/4	1736	24	1	2304	576	≥1
	1.4 - 20	5	2	7	12	16QAM	3/4	2152	24	1	2880	720	≥1
	3-20	6	2	7	12	16QAM	3/4	2600	24	1	3456	864	≥1
	3-20	8	2	7	12	16QAM	3/4	3496	24	1	4608	1152	≥1
	3-20	9	2	7	12	16QAM	3/4	3880	24	1	5184	1296	≥1
	3-20	10	2	7	12	16QAM	3/4	4264	24	1	5760	1440	≥1
	3-20	12	2	7	12	16QAM	3/4	5160	24	1	6912	1728	≥1
	5-20	15	2	7	12	16QAM	1/2	4264	24	1	8640	2160	≥1
	5-20	16	2	7	12	16QAM	1/2	4584	24	1	9216	2304	≥1
	5-20	18	2	7	12	16QAM	1/2	5160	24	1	10368	2592	≥1
	5-20	20	2	7	12	16QAM	1/3	4008	24	1	11520	2880	≥1
	5-20	24	2	7	12	16QAM	1/3	4776	24	1	13824	3456	≥1
	10-20	25	2	7	12	16QAM	1/3	4968	24	1	14400	3600	≥1
	10-20	27	2	7	12	16QAM	1/3	4776	24	1	15552	3888	≥1
	10-20	30	2	7	12	16QAM	3/4	12960	24	3	17280	4320	≥2
	10-20	32	2	7	12	16QAM	3/4	13536	24	3	18432	4608	≥2

	10-20	36	2	7	12	16QAM	3/4	15264	24	3	20736	5184	≥2
	10-20	40	2	7	12	16QAM	3/4	16992	24	3	23040	5760	≥2
	10-20	45	2	7	12	16QAM	3/4	19080	24	4	25920	6480	≥2
	10-20	48	2	7	12	16QAM	3/4	20616	24	4	27648	6912	≥2
	15 - 20	50	2	7	12	16QAM	3/4	21384	24	4	28800	7200	≥2
	15 - 20	54	2	7	12	16QAM	3/4	22920	24	4	31104	7776	≥2
	15 - 20	60	2	7	12	16QAM	2/3	23688	24	4	34560	8640	≥2
	15 - 20	64	2	7	12	16QAM	2/3	25456	24	4	36864	9216	≥2
	15 - 20	72	2	7	12	16QAM	1/2	20616	24	4	41472	10368	≥2
	20	75	2	7	12	16QAM	1/2	21384	24	4	43200	10800	≥2
	20	80	2	7	12	16QAM	1/2	22920	24	4	46080	11520	≥2
	20	81	2	7	12	16QAM	1/2	22920	24	4	46656	11664	≥2
	20	90	2	7	12	16QAM	2/5	20616	24	4	51840	12960	≥2
	20	96	2	7	12	16QAM	2/5	22152	24	4	55296	13824	≥2

NOTE 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)
NOTE 2: As per Table 4.2-2 in TS 36.211 [7]
NOTE 3: As per Table 4.2-1 in TS 36.211 [7]

A.2.2.2.3 64-QAM

Table A.2.2.2.3-1: Reference Channels for 64-QAM with partial RB allocation

Parameter	Ch BW	Allocated RBs	UL-DL Configuration (NOTE 2)	Special subframe configuration (NOTE 3)	DFT-OFDM Symbols per Sub-Frame	Mod'n	Target Coding rate	Payload size for Sub-Frame 2, 7	Trans-port block CRC	Number of code blocks per Sub-Frame (NOTE 1)	Total number of bits per Sub-Frame for Sub-Frame 2, 7	Total symbols per Sub-Frame for Sub-Frame 2, 7	UE Category (NOTE 4)
Unit	MHz							Bits	Bits		Bits		
	1.4 - 20	1	2	7	12	64QAM	3/4	616	24	1	864	144	5,8
	1.4 - 20	2	2	7	12	64QAM	3/4	1256	24	1	1728	288	5,8
	1.4 - 20	3	2	7	12	64QAM	3/4	1864	24	1	2592	432	5,8
	1.4 - 20	4	2	7	12	64QAM	3/4	2536	24	1	3456	576	5,8
	1.4 - 20	5	2	7	12	64QAM	3/4	3112	24	1	4320	720	5,8
	3-20	6	2	7	12	64QAM	3/4	3752	24	1	5184	864	5,8
	3-20	8	2	7	12	64QAM	3/4	5160	24	1	6912	1152	5,8
	3-20	9	2	7	12	64QAM	3/4	5736	24	1	7776	1296	5,8
	3-20	10	2	7	12	64QAM	3/4	6200	24	2	8640	1440	5,8
	3-20	12	2	7	12	64QAM	3/4	7480	24	2	10368	1728	5,8
	5-20	15	2	7	12	64QAM	3/4	9528	24	2	12960	2160	5,8
	5-20	16	2	7	12	64QAM	3/4	10296	24	2	13824	2304	5,8
	5-20	18	2	7	12	64QAM	3/4	11448	24	2	15552	2592	5,8
	5-20	20	2	7	12	64QAM	3/4	12576	24	3	17280	2880	5,8
	5-20	24	2	7	12	64QAM	3/4	15264	24	3	20736	3456	5,8
	10-20	25	2	7	12	64QAM	3/4	15840	24	3	21600	3600	5,8
	10-20	27	2	7	12	64QAM	3/4	16992	24	3	23328	3888	5,8
	10-20	30	2	7	12	64QAM	3/4	19080	24	4	25920	4320	5,8
	10-20	32	2	7	12	64QAM	3/4	20616	24	4	27648	4608	5,8
	10-20	36	2	7	12	64QAM	3/4	22920	24	4	31104	5184	5,8
	10-20	40	2	7	12	64QAM	3/4	25456	24	5	34560	5760	5,8
	10-20	45	2	7	12	64QAM	3/4	28336	24	5	38880	6480	5,8
	10-20	48	2	7	12	64QAM	3/4	30576	24	5	41472	6912	5,8
	15 - 20	50	2	7	12	64QAM	3/4	31704	24	6	43200	7200	5,8
	15 - 20	54	2	7	12	64QAM	3/4	34008	24	6	46656	7776	5,8
	15 - 20	60	2	7	12	64QAM	3/4	37888	24	7	51840	8640	5,8
	15 - 20	64	2	7	12	64QAM	3/4	40576	24	7	55296	9216	5,8
	1.4 - 20	1	2	7	12	64QAM	3/4	616	24	1	864	144	5,8
	1.4 - 20	2	2	7	12	64QAM	3/4	1256	24	1	1728	288	5,8
	1.4 - 20	3	2	7	12	64QAM	3/4	1864	24	1	2592	432	5,8
	1.4 - 20	4	2	7	12	64QAM	3/4	2536	24	1	3456	576	5,8

	1.4 - 20	5	2	7	12	64QAM	3/4	3112	24	1	4320	720	5,8
	3-20	6	2	7	12	64QAM	3/4	3752	24	1	5184	864	5,8
	3-20	8	2	7	12	64QAM	3/4	5160	24	1	6912	1152	5,8
	3-20	9	2	7	12	64QAM	3/4	5736	24	1	7776	1296	5,8
	3-20	10	2	7	12	64QAM	3/4	6200	24	2	8640	1440	5,8
	3-20	12	2	7	12	64QAM	3/4	7480	24	2	10368	1728	5,8
	5-20	15	2	7	12	64QAM	3/4	9528	24	2	12960	2160	5,8
	5-20	16	2	7	12	64QAM	3/4	10296	24	2	13824	2304	5,8
	5-20	18	2	7	12	64QAM	3/4	11448	24	2	15552	2592	5,8
	5-20	20	2	7	12	64QAM	3/4	12576	24	3	17280	2880	5,8
	5-20	24	2	7	12	64QAM	3/4	15264	24	3	20736	3456	5,8
	10-20	25	2	7	12	64QAM	3/4	15840	24	3	21600	3600	5,8
	10-20	27	2	7	12	64QAM	3/4	16992	24	3	23328	3888	5,8
	10-20	30	2	7	12	64QAM	3/4	19080	24	4	25920	4320	5,8
	15 - 20	50	2	7	12	64QAM	3/4	31704	24	6	43200	7200	5,8
	15 - 20	54	2	7	12	64QAM	3/4	34008	24	6	46656	7776	5,8
	15 - 20	60	2	7	12	64QAM	3/4	37888	24	7	51840	8640	5,8
	15 - 20	64	2	7	12	64QAM	3/4	40576	24	7	55296	9216	5,8
	15 - 20	72	2	7	12	64QAM	3/4	45352	24	8	62208	10368	5,8
	20	75	2	7	12	64QAM	3/4	46888	24	8	64800	10800	5,8
	20	80	2	7	12	64QAM	3/4	51024	24	9	69120	11520	5,8
	20	81	2	7	12	64QAM	3/4	51024	24	9	69984	11664	5,8
	20	90	2	7	12	64QAM	3/4	51024	24	9	77760	12960	5,8
	20	96	2	7	12	64QAM	3/4	61664	24	11	82944	13824	5,8

NOTE 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)
NOTE 2: As per Table 4.2-2 in TS 36.211 [7].
NOTE 3: As per Table 4.2-1 in TS 36.211 [7]
NOTE 4: If UE does not report UE UL category, then the applicability of reference channel is determined by UE category. If UE reports UE UL category, then the applicability of reference channel is determined by UE UL category

A.2.2.2.4 256 QAM

Table A.2.2.2.4-1: Reference Channels for 256 QAM with partial RB allocation

Parameter	Ch BW	Allocated RBs	UL-DL Configuration (NOTE 2)	Special Slot Configuration (NOTE 3)	DFT-OFDM Symbols per Sub-Frame	Mod'n	Target Coding rate	Payload size for Sub-Frame 2, 7	Trans-port block CRC	Number of code blocks per Sub-Frame (NOTE 1)	Total number of bits per Sub-Frame for Sub-Frame 2, 7	Total symbols per Sub-Frame for Sub-Frame 2, 7	UE UL Cateogry
Unit	MHz							Bits	Bits		Bits		
	1.4 - 20	1	2	7	12	256QAM	3/4	840	24	1	1152	144	≥ 15
	1.4 - 20	2	2	7	12	256QAM	3/4	1672	24	1	2304	288	≥ 15
	1.4 - 20	3	2	7	12	256QAM	3/4	2536	24	1	3456	432	≥ 15
	1.4 - 20	4	2	7	12	256QAM	3/4	3368	24	1	4608	576	≥ 15
	1.4 - 20	5	2	7	12	256QAM	3/4	4264	24	1	5760	720	≥ 15
	3-20	6	2	7	12	256QAM	3/4	5160	24	1	6912	864	≥ 15
	3-20	8	2	7	12	256QAM	3/4	6712	24	2	9216	1152	≥ 15
	3-20	9	2	7	12	256QAM	3/4	7736	24	2	10368	1296	≥ 15
	3-20	10	2	7	12	256QAM	3/4	8504	24	2	11520	1440	≥ 15
	3-20	12	2	7	12	256QAM	3/4	10296	24	2	13824	1728	≥ 15
	5-20	15	2	7	12	256QAM	3/4	12960	24	3	17280	2160	≥ 15
	5-20	16	2	7	12	256QAM	3/4	13536	24	3	18432	2304	≥ 15
	5-20	18	2	7	12	256QAM	3/4	15264	24	3	20736	2592	≥ 15
	5-20	20	2	7	12	256QAM	3/4	16992	24	3	23040	2880	≥ 15
	5-20	24	2	7	12	256QAM	3/4	20616	24	4	27648	3456	≥ 15
	10-20	25	2	7	12	256QAM	3/4	21384	24	4	28800	3600	≥ 15
	10-20	27	2	7	12	256QAM	3/4	22920	24	4	31104	3888	≥ 15
	10-20	30	2	7	12	256QAM	3/4	25456	24	5	34560	4320	≥ 15
	10-20	32	2	7	12	256QAM	3/4	27376	24	5	36864	4608	≥ 15

	10-20	36	2	7	12	256QAM	3/4	30576	24	6	41472	5184	≥ 15
	10-20	40	2	7	12	256QAM	3/4	34008	24	6	46080	5760	≥ 15
	10-20	45	2	7	12	256QAM	3/4	37888	24	7	51840	6480	≥ 15
	10-20	48	2	7	12	256QAM	3/4	40576	24	8	55296	6912	≥ 15
	15 - 20	50	2	7	12	256QAM	3/4	42368	24	8	57600	7200	≥ 15
	15 - 20	54	2	7	12	256QAM	3/4	46888	24	8	62208	7776	≥ 15
	15 - 20	60	2	7	12	256QAM	3/4	51024	24	9	69120	8640	≥ 15
	15 - 20	64	2	7	12	256QAM	3/4	55056	24	9	73728	9216	≥ 15
	15 - 20	72	2	7	12	256QAM	3/4	61664	24	11	82944	10368	≥ 15
	20	75	2	7	12	256QAM	3/4	63776	24	11	86400	10800	≥ 15
	20	80	2	7	12	256QAM	3/4	68808	24	12	92160	11520	≥ 15
	20	81	2	7	12	256QAM	3/4	68808	24	12	93312	11664	≥ 15
	20	90	2	7	12	256QAM	3/4	76208	24	13	103680	12960	≥ 15
	20	96	2	7	12	256QAM	3/4	81176	24	14	110592	13824	≥ 15
NOTE 1: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) NOTE 2: As per Table 4.2-2 in TS 36.211 [7] NOTE 3: As per Table 4.2-1 in TS 36.211 [7]													

A.3 DL reference measurement channels for E-UTRA

A.3.1 General

The number of available channel bits varies across the sub-frames due to PBCH and PSS/SSS overhead. The payload size per sub-frame is varied in order to keep the code rate constant throughout a frame.

Unless otherwise stated, no user data is scheduled on subframes #5 in order to facilitate the transmission of system information blocks (SIB).

The algorithm for determining the payload size A is as follows; given a desired coding rate R and radio block allocation N_{RB}

1. Calculate the number of channel bits N_{ch} that can be transmitted during the first transmission of a given sub-frame.

2. Find A such that the resulting coding rate is as close to R as possible, that is,

   ,

   subject to

   a) A is a valid TB size according to clause 7.1.7 of TS 36.213 [6] assuming an allocation of N_{RB} resource blocks.

   b) C is the number of Code Blocks calculated according to clause 5.1.2 of TS 36.212 [5].

3. If there is more than one A that minimizes the equation above, then the larger value is chosen per default and the chosen code rate should not exceed 0.93.

4. For TDD, the measurement channel is based on DL/UL configuration ratio of 3DL+DwPTS (10 OFDM symbol SSF7): 1UL

A.3.1.1 QPSK

Table A.3.1.1-1: Fixed Reference Channel for Receiver Requirements (TDD)

Parameter	Unit	Value					
Channel Bandwidth	MHz	1.4	3	5	10	15	20

Allocated resource blocks		6	15	25	50	75	100
Uplink-Downlink Configuration (NOTE 5)		2	2	2	2	2	2
Special subframe configuration (NOTE 6)		7	7	7	7	7	7
Allocated subframes per Radio Frame (D+S)		3	3+2	3+2	3+2	3+2	3+2
Number of HARQ Processes	Processes	7	7	7	7	7	7
Maximum number of HARQ transmission		1	1	1	1	1	1
Modulation		QPSK	QPSK	QPSK	QPSK	QPSK	QPSK
Target coding rate		1/3	1/3	1/3	1/3	1/3	1/3
Information Bit Payload per Sub-Frame	Bits						
For Sub-Frame 3, 4, 8, 9		408	1320	2216	4392	6712	8760
For Sub-Frame 1, 6		N/A	776	1288	2664	4008	5352
For Sub-Frame 5		N/A	N/A	N/A	N/A	N/A	N/A
For Sub-Frame 0		208	1064	1800	4392	6712	8760
Transport block CRC	Bits	24	24	24	24	24	24
Number of Code Blocks per Sub-Frame (NOTE 4)							
For Sub-Frame 3, 4, 8, 9		1	1	1	1	2	2
For Sub-Frame 1, 6		N/A	1	1	1	1	1
For Sub-Frame 5		N/A	N/A	N/A	N/A	N/A	N/A
For Sub-Frame 0		1	1	1	1	2	2
Binary Channel Bits Per Sub-Frame	Bits						
For Sub-Frame 3, 4, 8, 9		1368	3780	6300	13800	20700	27600
For Sub-Frame 1, 6		N/A	2616	4456	9056	13656	18256
For Sub-Frame 5		N/A	N/A	N/A	N/A	N/A	N/A
For Sub-Frame 0		672	3084	5604	13104	20004	26904
Max. Throughput averaged over 1 frame	kbps	102.4	564	932	1965.6	3007.2	3970.4
UE Category		≥ 1	≥ 1	≥ 1	≥ 1	≥ 1	≥ 1

NOTE 1: For normal subframes(0,3,4,5,8,9), 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz. For special subframe (1&6), only 2 OFDM symbols are allocated to PDCCH for all BWs.
NOTE 2: For 1.4MHz, no data shall be scheduled on special subframes(1&6) to avoid problems with insufficient PDCCH performance
NOTE 3: Reference signal, Synchronization signals and PBCH allocated as per TS 36.211 [7]
NOTE 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
NOTE 5: As per Table 4.2-2 in TS 36.211 [7]
NOTE 6: As per Table 4.2-1 in TS 36.211 [7]

A.3.1.2 64-QAM

Table A.3.1.2-1: Fixed Reference Channel for Maximum input level for UE Categories ≥ 3 (TDD)

Parameter	Unit	Value					
Channel bandwidth	MHz	1.4	3	5	10	15	20
Allocated resource blocks		6	15	25	50	75	100
Subcarriers per resource block		12	12	12	12	12	12
Uplink-Downlink Configuration (NOTE 5)		2	2	2	2	2	2
Special subframe configuration (NOTE 6)		7	7	7	7	7	7
Allocated subframes per Radio Frame		2	3+2	3+2	3+2	3+2	3+2
Modulation		64QAM	64QAM	64QAM	64QAM	64QAM	64QAM
Target Coding Rate		¾	¾	¾	¾	¾	¾
Number of HARQ Processes	Processes	7	7	7	7	7	7
Maximum number of HARQ transmissions		1	1	1	1	1	1
Information Bit Payload per Sub-Frame							
For Sub-Frames 3, 4, 8, 9	Bits	2984	8504	14112	30576	46888	61664
For Sub-Frames 1,6	Bits	N/A	5544	9528	19848	30576	40576
For Sub-Frame 5	Bits	N/A	N/A	N/A	N/A	N/A	N/A
For Sub-Frame 0	Bits	N/A	6968	12576	30576	45352	61664

Transport block CRC	Bits	24	24	24	24	24	24
Number of Code Blocks per Sub-Frame (NOTE 4)							
For Sub-Frames 3, 4, 8, 9		1	2	3	5	8	11
For Sub-Frames 1,6		N/A	2	2	4	6	8
For Sub-Frame 5		N/A	N/A	N/A	N/A	N/A	N/A
For Sub-Frame 0		N/A	2	3	5	8	11
Binary Channel Bits per Sub-Frame							
For Sub-Frames 3, 4, 8, 9	Bits	4104	11340	18900	41400	62100	82800
For Sub-Frames 1,6		N/A	7848	13368	27168	40968	54768
For Sub-Frame 5	Bits	N/A	N/A	N/A	N/A	N/A	N/A
For Sub-Frame 0	Bits	N/A	9252	16812	39312	60012	80712
Max. Throughput averaged over 1 frame	kbps	596.8	3791.2	6369.6	13910	20945	27877

NOTE 1: For normal subframes(0,3,4,5,8,9), 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz. For special subframe (1&6), only 2 OFDM symbols are allocated to PDCCH for all BWs.
NOTE 2: For 1.4MHz, no data shall be scheduled on special subframes(1&6) to avoid problems with insufficient PDCCH performance.
NOTE 3: Reference signal, Synchronization signals and PBCH allocated as per TS 36.211 [7].
NOTE 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
NOTE 5: As per Table 4.2-2 in TS 36.211 [7].
NOTE 6: As per Table 4.2-1 in TS 36.211 [7]

A.3.1.3 256-QAM

Table A.3.1.3-1: Fixed Reference Channel for Maximum input level for UE Categories 11/12 and UE DL categories ≥ 11 (TDD)

Parameter	Unit	Value					
Channel bandwidth	MHz	1.4	3	5	10	15	20
Allocated resource blocks		6	15	25	50	75	100
Subcarriers per resource block		12	12	12	12	12	12
Uplink-Downlink Configuration (NOTE 5)		2	2	2	2	2	2
Special subframe configuration (NOTE 6)		7	7	7	7	7	7
Allocated subframes per Radio Frame		2	3+2	3+2	3+2	3+2	3+2
Modulation		256QAM	256QAM	256QAM	256QAM	256QAM	256QAM
Target Coding Rate		4/5	4/5	4/5	4/5	4/5	4/5
Number of HARQ Processes	Processes	7	7	7	7	7	7
Maximum number of HARQ transmissions		1	1	1	1	1	1
Information Bit Payload per Sub-Frame							
For Sub-Frames 3,4,8,9	Bits	4392	12216	19848	42368	63776	84760
For Sub-Frames 1,6	Bits	N/A	10464	17824	36224	54624	73024
For Sub-Frame 5	Bits	N/A	N/A	N/A	N/A	N/A	N/A
For Sub-Frame 0	Bits	N/A	9912	17568	42368	63776	84760
Transport block CRC	Bits	24	24	24	24	24	24
Number of Code Blocks per Sub-Frame (NOTE 4)							
For Sub-Frames 3,4,8,9		1	2	4	7	11	14
For Sub-Frames 1,6		N/A	2	3	6	9	13
For Sub-Frame 5		N/A	N/A	N/A	N/A	N/A	N/A
For Sub-Frame 0		N/A	2	3	7	11	14
Binary Channel Bits per Sub-Frame							
For Sub-Frames 3,4,8,9	Bits	5472	15120	25200	55200	82800	110400
For Sub-Frames 1,6		N/A	8248	13536	27376	40576	55056
For Sub-Frame 5	Bits	N/A	N/A	N/A	N/A	N/A	N/A
For Sub-Frame 0	Bits	N/A	12336	22416	52416	80016	107616
Max. Throughput averaged over 1 frame	kbps	878.4	5570.4	9240	20049.6	30144	40503.2

NOTE 1:	For normal subframes(0,3,4,5,8,9), 2 symbols allocated to PDCCH for 20 MHz, 15 MHz and 10 MHz channel BW; 3 symbols allocated to PDCCH for 5 MHz and 3 MHz; 4 symbols allocated to PDCCH for 1.4 MHz. For special subframe (1&6), only 2 OFDM symbols are allocated to PDCCH for all BWs.
NOTE 2:	For 1.4MHz, no data shall be scheduled on special subframes(1&6) to avoid problems with insufficient PDCCH performance.
NOTE 3:	Reference signal, Synchronization signals and PBCH allocated as per TS 36.211 [7].
NOTE 4:	If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
NOTE 5:	As per Table 4.2-2 in TS 36.211 [7].
NOTE 6:	As per Table 4.2-1 in TS 36.211 [7]

Annex B: Void

Annex C: Void

Annex D: Void

Annex E: Void

Annex F: Void

Annex G: Void

Annex H (normative): Modified MPR behavior

Annex I (normative): Dual uplink interferer

UE is mandated to support operation in dual and triple uplink mode for EN-DC configuration in NR FR1 listed in Table 5.5B.2-1, Table 5.5B.3-1, and Table 5.5B.4.1-1 and indicated by column single uplink allowed, Table 7.3B.2.3.5.1-1, Table 7.3B.2.3.5.2-0, Table 7.3B.2.3.5.2-1 or NE-DC configuration in NR FR1 listed in Table 5.5B.4a.1-1 and indicated by column single uplink allowed if the intermodulation products caused by the dual uplink operation do not interfere with its own primary downlink transmission channel bandwidth of PCell or PSCell. For intermodulation products falling into any secondary downlink channel bandwidth, UE single UL capability is not considered.

Formula for determining if the EN-DC in NR FR1 configuration with dual uplink operation interferes with its own downlink reception.

Interference bandwidth: IBW = |a| * CBW1 + |b| * CBW2

- |a| + |b| = 2 (or 3)
- CBW1 and CBW2 are the transmission bandwidth configurations of the UL channels

Center frequency of IBW: fIBW = |a * f1 + b * f2|

- f1 and f2 are center frequency of the transmission bandwidth configurations of each UL channel

The range of IMD 2 (or 3): [fIBW – IBW/2, fIBW + IBW/2]

NOTE 1: UE shall be able to apply operations which are configured by RRC reconfiguration and corresponding HARQ timing on the transmission bandwidth.

NOTE 2: For identified difficult band combination, during two adjacent RRC reconfiguration, the changing of transmission bandwidth should not introduce IM2 and IM3, which will result in UE changing from 2Tx to 1Tx. Otherwise, UE behavior is not specified.

For DC_3A_n3A intra-band non-contiguous EN-DC combination, only single switched UL is supported in Rel-15.

Annex J: Void

Annex K: Void

Annex L (informative): Change history

Change history							
Date	Meeting	TDoc	CR	Rev	Cat	Subject/Comment	New version
2017-08	RAN4#84					Initial Skeleton	0.0.1
2017-11	RAN4#84 Bis	R4-1711980				Number TPs from editors	0.1.0
2017-12	RAN4#85	R4-1713807				Approved TPs in RAN4#85 R4-1714444, CA BW classes, TP, Ericsson R4-1714170, How to list DC configurations into TS 38.101-3, Nokia R4-1714530, TP on introducing operating bands for NR-LTE DC including SUL band combinations in 38.101-3 , Qualcomm R4-1714098, TP to TS 38.101-3: UE RF requirements for non-standalone SUL, Huawei R4-1713206, TP on general parts for 38.101-3 NR interwork, Ericsson R4-1714443, TP to TS 38.101-3: On dual uplink operation for EN-DC in NR FR1 and single uplink, Nokia R4-1714450, TP to 38.101-3: maximum output power and unwanted emissions for EN-DC, Ericsson R4-1714346, TP to 38.101-3: REFSENS for intra-band EN-DC, Ericsson R4-1714345, TP for TS 36.101-3: clause 7 receiver requirements, Huawei Band list according to R4-1714542, List of bands and band combinations to be introduced into RAN4 NR core requirements by December 2017, RAN4 Chairmen	0.2.0
2017-12	RAN4#85	R4-1714571				Further corrections after email review	0.3.0
2017-12	RAN#78	RP-172477				v1.0.0 submitted for plenary approval. Contents same as 0.3.0	1.0.0
2017-12	RAN#78					Approved by plenary – Rel-15 spec under change control	15.0.0
2018-03	RAN#79	RP-180264	0005		F	Implementation of endorsed CRs to 38.101-3 Endorsed draft CR F: R4-1801267, Draft CR on UE RF requirements for SUL in TS 38.101-3, Huawei B: R4-1801111, Draft CR for completed LTE 1CC + NR 1band for TS 38.101-3, NTT DOCOMO, INC. B: R4-1800716, Draft CR for introduction of completed band combinations from 37.863-03-01 into 38.101-3, Ericsson B: R4-1800063, Draft CR for completed EN-DC of LTE 4CC + NR 1band for TS 38.101-3, Nokia B: R4-1800717, Draft CR for introduction of completed band combinations from 37.865-01-01 into 38.101-3, Ericsson F: R4-1800049, Modification for TS38.101-3, CATT F: R4-1800287, 38.101-3 DC_(n)71B draft CR for section 6.2.4.1 - A-MPR for intra-band EN-DC - NS value, T-Mobile USA Inc. F: R4-1800288, 38.101-3 DC_(n)71B draft CR for section 7.3.3 Reference sensitivity for DC_(n)71B - MSD values, T-Mobile USA Inc. F: R4-1801139 Draft CR to 38.101-3: MSD for inter-band EN-DC, Ericsson	15.1.0
2018-06	RAN#80	RP-181374	0013	1	F	CR to TS 38.101-3: Implementation of endorsed draft CRs from RAN4 #87 **Missing figures (Figure 6.3B.1.1-1, Figure 6.3B.1.1-2, Figure 6.3B.1.1-3 and Figure 6.3B.1.1-4) from the endorsed draftCR (R4-1807235) were added during the CR implementation.**	15.2.0
2018-09	RAN#81	RP-182129	0020	2	F	Big CR for 38.101-3 Draft CRs from RAN4#88: R4-1809960 Draft CR to TS 38.101-3: to introduce new NR	15.3.0

						inter-band DC band combinations Samsung,KDDI,SKT,KT,LGU+ R4-1809991 CR to 38.101-3:Corrections on UE coexistence table for Table 6.5B.3.3.1-1 MediaTek Inc. R4-1810054 Pcmax for Rel-15 inter-band EN-DC for FR1 and NR in FR2 InterDigital, Inc. R4-1810111 Single UL allowed corrections for DC_28A-n51A EN-DC in 38.101-3 Skyworks Solutions Inc. R4-1810125 Draft CR to 38.101-3 Single UL allowed corrections for DC_28A_51A EN-DC Skyworks Solutions Inc. R4-1810128 Draft CR to 38.101-3 Single UL allowed corrections for EN-DC operation in NR FR1 (two bands) Skyworks Solutions Inc. R4-1810167 TP for TR 37.863-01-01: MSD for DC_5A_n78A due to the 4th harmonic MediaTek Inc. R4-1810410 Draft CR to 38.101-3: Corrections on symbols and abbreviations in section 3 ZTE Corporation R4-1810417 Correction to DC_(n)71B MSD definition Nokia R4-1810433 Correction on EN-DC 8A_n79A SoftBank Corp.,ZTE R4-1810476 Draft CR to TS 38.101-3 correction for DC_3_n3-n77, DC_3_n3-n78 CHTTL R4-1810976 Annex lettering change for 38.101-3 Qualcomm Incorporated R4-1811461 Clarification and corrections of EN-DC REFSENS exceptions requirement Nokia, Nokia Shanghai Bell R4-1811462 Correction to DC_(n)71B scs restriction for NR Nokia R4-1811466 EN DC_41-79 CATT R4-1811467 Draft CR TS 38.101-3 Corrections to Single UL Allowed Criteria for Mid-Band EN-DC in FR1 Skyworks Solutions Inc. R4-1811484 Pcmax for inter-band EN-DC FR1 draft CR InterDigital, Inc. R4-1811525 Draft CR TS 38.101-3 on missing requirements for FR1 EN-DC Skyworks Solutions, Inc. R4-1811542 Draft CR to 38.101-3 on correction on some errors Huawei, HiSilicon R4-1811796 Draft CR to 38.101-3 Corrections to Single UL allowed criteria for EN-DC Skyworks Solutions Inc. R4-1811800 DRAFT CR for PCmax FR2 correction Qualcomm Incorporated R4-1811810 Draft CR TS 38.101-3: Corrections for B41/ n41 SPRINT Corporation	
2018-12	RAN#82	RP-182359	0030		F	Endorced draft CRs from RAN4#88Bis : R4-1812057, Introduction of Intra-band contiguous EN-DC bandwidth classes, Nokia R4-1812290 Draft CR on MSD for EN-DC including Band 66 and n78 Huawei, HiSilicon R4-1812293 Draft CR on switching time mask for EN-DC Huawei, HiSilicon R4-1812298 Draft CR to TS 38.101-3: to add missing requirements for inter-band CA between FR1 and FR2. Samsung R4-1812360 Draft CR to 38.101-3: Corrrection to UL configuration for EN-DC reference sensitivity exceptions Skyworks Solutions Inc. R4-1812361 Draft CR to 38.101-3: NR uplink DFT-S-OFDM waveforms for EN-DC reference sensitivity Skyworks Solutions Inc. R4-1812362 Draft CR to 38.101-3: Editorial and RB allocation corrections to table 7.3B.2.3.4-2 Skyworks Solutions Inc. R4-1812363 Draft CR to 38.101-3: Single UL allowed operation corrections in clause 7.3B.2.3.5 Skyworks Solutions Inc. R4-1812404 Non-contiguous intra-band EN-DC emission requirements Qualcomm Incorporated R4-1812410 Correction on REFSENS exception for EN-DC 41A-n77A/n78A SoftBank Corp. R4-1812670 Correction on REFSENS exceptions of DC_5A-7A_n78A to TS 38.101-3 LG Uplus R4-1813471 draftCR on applicability of TDD configuratiin for CA in TS 38.101-3 Huawei R4-1813796 Draft CR for 38.101-3: Intra-band Pcmax for Type 2 UEs Sprint Corporation R4-1813816 Renaming of DC_(n)71B into	15.4.0

						DC_(n)71AA Nokia R4-1813817 Correction to EN-DC operating bands and configurations Nokia R4-1813818 Draft CR on correction REFSENs exceptions due to dual uplink operation for inter-band EN-DC to TS 38.101-3 Samsung R4-1813822 Draft CR for 38.101-3: Single UL allowed criteria in Annex I Vodafone España SA R4-1814157 Draft CR for UE-to-UE coexistence requirements for intra-band EN-DC in TS38.101-3 LG Electronics France R4-1814167 Draft CR on Single UL for some EN-DC combinations Huawei Endorsed draft CRs from Ran4#89: R4-1815952 dCR on TS38.101-3 merging draft CRs from RAN4#(88Bis) Qualcomm IncorporatedR4-1814803 Draft CR on editorial error for EN-DC band combinations to TS 38.101-3 Huawei, HiSilicon R4-1815802 draft CR editorial correction in 38.101-3 Ericsson R4-1814425 Simplification of requirements for EN-DC configuration including FR2 NTT DOCOMO, INC. R4-1814512 Draft CR to TS38.101-3_Corrections on MSD requirments for EN-DC combinations of band 8 and n77 n78(Section 7.3B.2.3.1) ZTE Corporation R4-1814938 Draft CR to 38.101-3 on operating bands for CA and DC ZTE Corporation Zhifeng Ma R4-1814976 Correction for Maximum output power for inter-band EN-DC (two bands)Nokia, Nokia Shanghai Bell R4-1814977 Correction for ?TIB,c for EN-DC Nokia, Nokia Shanghai Bell R4-1814978 MPR and A-MPR for interband EN-DC Nokia, Nokia Shanghai Bell R4-1814980 Correction for intra-band EN-DC bandwidth class Nokia, Nokia Shanghai Bell R4-1815065 draft CR for adding missing transmit singnal quality for inter band EN-DC for TS 38.101-3 NTT DOCOMO, INC. R4-1815811 draft Rel-15 CR to 38.101-3 to correct n260 BW class Ericsson, AT&T R4-1815865 Draft CR for 38.101-3 Intra-band EN-DC nominal carrier spacing for 30 kHz raster SPRINT Corporation R4-1815973 Draft CR to 38.101-3 rel. 15 to fix MSD issues for higher order EN-DC combinations R4-1816227 Draft CR on Power Class for inter band EN-DC within FR1 OPPO R4-1816233 Receiver requirements for intra-band EN-DC Qualcomm Incorporated R4-1816621 Introduction of maxUplinkDutyCycle to ENDC HPUE in FR1 OPPO R4-1816638 Pcmax computation and evaluation for inter band ENDC Qualcomm R4-1816178 Draft CR for correction for missing agreed DC combinations in Rel-15 for TS 38.101-3 NTT DOCOMO, INC. R4-1816197 Draft CR to TS38.101-3_Clarifications on MSD and UL configuration tables for EN-DC ZTE Corporation R4-1816198 Simplification of EN-DC and CA between FR1 and FR2 UE to UE co-ex table by adopting CA band approach Nokia, Nokia Shanghai Bell R4-1816202 Correction to interband EN-DC OOBE emission requirements Nokia, Nokia Shanghai Bell R4-1816203 Receiver requirements for interband EN-DC Nokia, Nokia Shanghai Bell R4-1816207 Draft CR to 38.101-3 rel. 15 to fix MPR issue Apple GmbH R4-1816224 Draft CR for 38.101-3 NS_04 applicability for intra-band EN-DC SPRINT Corporation R4-1816231 Draft CR on output power dynamic for DC OPPO R4-1816237 Correction for Intra-band contiguous EN-DC A-MPR definition Nokia, Nokia Shanghai Bell R4-1816246 Draft CR to TS38.101-3: Corrections on TS for MSD calculations based on ENDC bands combination including of bands 1,3,8, n77, and n78 MediaTek Inc. R4-1816247 Draft CR 38-101-3 Corrections for EN-DC Single Uplink allowed Operation Skyworks Solutions Inc. R4-1816250 draft CR for adding note about the fallback of	

						EN-DC in Applicability of minimum requirements for TS 38.101-3 NTT DOCOMO, INC. R4-1816608 Draft CR on LTE RMC for TDD EN-DC UE RF Tests Qualcomm Incorporated R4-1816613 Draft CR for reducing AMPR for DC_(n)71AA without Dynamic Power Sharing " Motorola Mobility, T-Mobile"	
2018-12	RAN#82	RP-182773	0033	1	F	Completion of configured maximum output power for intra-band contiguous EN-DC	15.4.0
2018-12	RAN#82	RP-182774	0034	1	F	Configured maximum output power for intra-band non-contiguous EN-DC	15.4.0
2019-03	RAN#83	RP-190403	0035		F	CR to TS 38.101-3: Implementation of endorsed draft CRs from RAN4#90 Endorced draft CRs from RAN4#90 R4-1900034, Editorial corrections for 38.101-3, Qualcomm Incorporated R4-1900460, Draft CR to TS38.101-3_corrections on MSD, ZTE Corporation R4-1900461, Draft CR to TS38.101-3_inter-band NR DC between FR1 and FR2, ZTE Corporation R4-1900524, Draft CR to TS 38.101-3 on inter-band CA & inter-band EN-DC configurations, ZTE Corporation R4-1900529, Draft CR to TS 38.101-3 on Single Uplink Allowed for EN-DC combinations of order 3 or higher, ZTE Corporation R4-1900726, Editorial corrections to delta Tib for EN-DC, Rohde & Schwarz R4-1901359, draft CR for correction for missing operating band for EN-DC, NTT DOCOMO INC. R4-1901428, draft CR to make editorial corrections in 38-101-3 Rel-15, Ericsson R4-1901848, Draft CR for 38.101-3: Addition of default power class, Sprint Corporation R4-1901850, Draft CR for 38.101-3: Intra-band Pcmax P_EN-DC_Total for non-DPS UEs, Sprint Corporation R4-1901851, Draft CR for 38.101-3: Intra-band Pcmax Editorial corrections, Sprint Corporation R4-1901874, Guardband for harmonic exception to reference sensitivity, Qualcomm Incorporated R4-1901878, Non-simultaneous Tx/Rx for TDD intra-band EN-DC, Qualcomm Incorporated R4-1901890, A-MPR for DC_(n)71AA without Dynamic Power Sharing, Motorola Mobility France S.A.S R4-1901926, Draft CR to 38.101-3 to clarify ACS2 wanted level, Qualcomm Incorporated R4-1901997, draft_CR TS 38.101-3 type 2 UE DC_(n)41 and DC_41_n41 NS04 AMPR correction, Skyworks Solutions Inc. R4-1902002, Draft CR to 38.101-3 on DC_n41-41 – B40 coexistence , Qualcomm Incorporated R4-1902154, Draft CR to TS38.101-3_clean up on inter-band CA between FR1 and FR2, ZTE Corporation R4-1902155, Draft CR for TS 38.101-3: Corrections to Table 7.3B. 2.3.5.1-1 for reference sensitivity exceptions (two bands), MediaTek Inc. R4-1902156, draftCR corrections for TS 38.101-3, Huawei R4-1902157, CR on intraband ENDC channel configurations, Intel Corporation R4-1902160, Draft CR on some errors to TS 38.101-3, Huawei R4-1902161, CR to 38.101-3 to clarify non-simultaneous RXTX capability for co-bands, Qualcomm Incorporated R4-1902163, Draft CR to 38.101-3 to clarify DL carrier levels for bands in close frequency proximity, Qualcomm Incorporated R4-1902164, Draft CR to reflect agreed MSD analysis of DC_25A-n41A for TS 38.101-3, MediaTek Inc. R4-1902169, draft CR for inter-band EN-DC Pcmax, Huawei R4-1902172, Draft CR ACLR for NC intra-band EN-DC, Skyworks Solutions Inc. R4-1902176, Draft CR for 38.101-3 modification of requirements for intra-band non-contiguous EN-DC SEM, Huawei R4-1902179, draft CR for introduction of Tx IM for Inter-band EN-DC in TS38.101-3, NTT DOCOMO, INC. R4-1902182, Clarification for OOBE boundary for intra-band contiguous and non-contiguous EN-DC, vivo R4-1902195, draft_CR TS 38.101-3 Footnote correction in Table 7.3B.2.3.1-2, Skyworks Solutions Inc. R4-1902232, Draft CR on SUL band combinations to TS 38.101-3, Huawei	15.5.0

						R4-1902478, Addition of power class 2 EN-DC ACLR requirement, Nokia R4-1902481, draftCR on inter-band EN-DC Rx requirement for TS 38.101-3, Huawei R4-1902486, Draft CR for 38.101-3 modification of requirements for network signalled value NS_04, Huawei R4-1902496, Draft CR for TS 38.101-3: Switching time for intra-band EN-DC upon dual PA UE capability, Huawei R4-1902500, Draft CR for 38.101-3: adding MPR for intra-band ENDC,Skyworks Solutions Inc R4-1902660, Introduction of modified MPR for 38.101-3, Nokia Editorial changes after RAN#83 To align the annex numbering with other specifications (TS 38.101-x series), 'Modified MPR behavior' was moved to annex H.	
2019-06	RAN#84	RP-191240	0041		F	CR to TS 38.101-3: Implementation of endorsed draft CRs from RAN4#90bis and RAN4#91 Endorced draft CRs from RAN4#90Bis R4-1902829, Draft CR for 38.101-3 editoral correction for editorial correction for intra-band contiguous EN-DC uplink configuration when Rx requirements are measured, Huawei R4-1903074 Draft CR to 38.101-3 rel. 15 to fix missing SUO note Apple Inc. R4-1903090 Pcmax for Rel-15 intra-band EN-DC within FR1 wrong references - fixes InterDigital Communications R4-1903150 Draft CR to TS 38.101-3_Spurious emission and Tx IM for inter-band CA between FR1 and FR2 ZTE Corporation R4-1903302 Draft CR to TS 38.101-3 correction for the DC_3_n3 delta R IBNC table CHTTL R4-1903426 draft CR for 38.101-3: Reflect the agreed MSD for DC_5_n78 China Telecom R4-1903515 Removal of reference sensitivity exception due to close proximity of bands for EN-DC in NR FR1 clause Nokia R4-1903958 Completion of defintions of EN-DC configured power Ericsson R4-1904639 Draft CR to 38.101-3 on DC_n41-41 – B40 coexistence, Qualcomm Incorporated R4-1904934 Harmonization of reference sensitivity level for DC clause Nokia R4-1904935 Change description 4.2(e) in Applicability of minimum requirements for TS 38.101-3 vivo R4-1904945 Draft CR to TS38.101-3_adding some exclusion frequencies for SEM and spurious emission for EN-DC ZTE Corporation R4-1904946 Draft CR to TS 38.101-3 on some minor corrections ZTE Corporation R4-1904951 Draft CR for 38.101-3 intra-band EN-DC AMPR Huawei R4-1904953 Draft CR for 38.101-3: NS_04 A-MPR power class relationship clarification Sprint Corporation R4-1904959 Draft CR on UE to UE coexistence for TS 38.101-3 Huawei R4-1904988 Draft CR to 38.101-1. Clarify EN-DC category for requirements of carrier imbalance Qualcomm Incorporated R4-1904995 draft CR to 38.101-3 Configured output power for inter-band EN-DC including both FR1 and FR2 Intel Corporation R4-1905085 Draft CR for TR 38.101-3 NE-DC RF requirement Huawei R4-1904925 Draft CR for improving EN-DC configuration tables in TS38.101-3 CATT Endorced draft CRs from RAN4#91 R4-1905628 Draft CR to TS38.101-3_Frequency error for intra-band for EN-DC ZTE Corporation R4-1905629 Draft CR to TS 38.101-3_removal of the reference sensitivity exception for NR CA between FR1 and FR2 ZTE Corporation R4-1905767 draft CR to 38.101-3 Correction ot DeltaTIB,c in configured output power for EN-DC Intel Corporation R4-1905774 Draft CR to TS38.101-3 Correction to intra-band and inter-band EN-DC Pcmax Intel Corporation R4-1905793 CR for TS 38.101-3 (Rel-15): Support of n257D-F for DC_1-42_n257 and DC_3-42_n257 SoftBank Corp. R4-1905799 Correction of LTE anchor condition to Spurious response for EN-DC Anritsu Corporation R4-1907057 Draft CR for 38.101-3: Further UE coexistence table clean-up Sprint Corporation	15.6.0

						R4-1907063 Draft CR for 38.101-3: Global replacement of LTE with E-UTRA Sprint Corporation R4-1907136 Draft CR to 38.101-3 rel. 15 to fix missing Exceptions for Out-of-band Blocking Apple R4-1907137 Draft CR to 38.101-3 rel. 15 to fix missing SUO note Apple R4-1907181 Draft CR for 38.101-3: Removal of unnecessary ACLR notes Sprint Corporation R4-1907422 Draft CR for TS 38.101-1 Correction of channel bandwidth set for NR CA Huawei R4-1907424 Draft CR for clarification of note for B42_n77 and B42_n78 NTT DOCOMO, INC. R4-1907425 DraftCR TS 38.101-3 Corrections to Intra-band ENDC MPR text Skyworks Solutions Inc. R4-1907426 Definition of BCS support in inter-band EN-DC mode Qualcomm Incorporated R4-1907448 Correction to EN-DC spurious emissions ROHDE & SCHWARZ R4-1907476 draft CR for TS 38.101-3 intra-band EN-DC Pcmax Huawei R4-1907482 Correction of RefSens exceptions due to UL harmonic interference for EN-DC in 38.101-3 vivo R4-1907483 [Rx]Draft CR for 38.101-3 defining Reference sensitivity for intra-band non-contiguous, Huawei R4-1907485 Corrections to MPR/A-MPR and additional requirements for intra-band EN-DC Ericsson R4-1907489 Draft CR to 38.101-3. Revise MSD for DC_20A-n8A Qualcomm Incorporated R4-1907492 Modification of reference sensitivity and general spurious emissions in 38.101-3 Qualcomm Incorporated R4-1907594 draft CR of modification on reference for inter-band EN-DC including FR2 for TS 38.101-3 NTT DOCOMO INC. R4-1907808 Draft CR to 38.101-3 NE-DC introducation InterDigital Communications	
2019-12	RAN#86	RP-192049	0063		F	CR to TS 38.101-3: Implementation of endorsed draft CRs from RAN4#92 (Rel-15) R4-1907957 Reference sensitivity for intra-band EN-DC with single uplink Qualcomm Incorporated R4-1908028 Draft CR for handling of fallbacks for combined contiguous and non-contiguous CA in FR2 R4-1908162 dCR to 38.101-3: EN-DC REFSENS editorial updates Qualcomm Incorporated R4-1908376 Draft CR for TS38.101-3, Editorial corrections CATT R4-1908434 Further correction of RefSens exceptions due to UL harmonic interference for EN-DC in 38.101-3vivo R4-1908692 Editorial correction on SUO mark for BC 3+78 Apple Inc. R4-1908959 Draft CR for 38.101-3 adding note for spurious emission band UE co-existence Huawei, HiSilicon R4-1908960 Draft CR for 38.101-3 clean up for inter-band NE-DC Huawei, HiSilicon R4-1909065 Draft CR for efitorial correction on EN-DC of 4LTE bands + 1NR band for Rel-15 TS 38.101-3 NTT DOCOMO, INC. R4-1909354 Draft CR to TS 38.101-3 on UE additional maximum output power reduction for EN-DC ZTE Corporation R4-1909357 Draft CR to TS 38.101-3 on UE maximum output power for DC ZTE Corporation R4-1909798 Correction of DC configurations in tables chapter 5 Ericsson R4-1909903 Draft CR for 38.101-3: Correction of Note 2 in Table 5.5B.2-1 Sprint, Ericsson, Google R4-1909956 dCR to 38.101-3 EN-DC RX Out-of-Band Blocking for Inter-bands treated as Intra-band Qualcomm Inc. R4-1909977 draftCR to 38.101-3 for missing Harmonic Mixing MSD fo DC_11A_n79A like DC 21A n79A Qualcomm Inc. R4-1910236 Draft CR of modification on reference to 38.101-2 requirements for inter-band EN-DC with FR2 for TS 38.101-3 NTT DOCOMO, INC. R4-1910245 draft CR to correct band of band26 related EN-DC KDDI R4-1910246 Draft CR to add simultaneous RX/TX capability requirements in R15 TS 38.101-3 CMCC R4-1910248 Draft CR to TS 38.101-3 for DC_38_n78 mandatory simultaneous Rx/Tx capability Vodafone	15.7.0

						R4-1910249 Draft CR for handling of EN-DC BCS for TS 38.101-3 NTT DOCOMO, INC. R4-1910250 draftCR to 38.101-3 for missing IMD MSD for shared bands that already have MSD Qualcomm Inc. R4-1910251 Draft CR for 38.101-3: non-contiguous resource allocation Huawei, HiSilicon R4-1910255 Draft Minor Corrections to 38.101-3 clause 6.2B.4.1.1 Motorola Mobility, TMobile USA R4-1910256 draft CR for 38.101-3 specifying occupied bandwidth for intra-band non-contiguous EN-DC Huawei, HiSilicon R4-1910257 Draft CR to TS 38.101-3: transmit intermodulation for intra-band ENDC ZTE Corporation R4-1910268 Draft CR for TS 38.101-3: Corrections to reference sensitivity exceptions due to receiver harmonic mixing and cross band isolation for ENDC in NR FR1 MediaTek Inc. R4-1910317 Draft CR for TS 38.101-3: Additional out-of-band blocking exceptions for inter-band EN-DC MediaTek Inc. R4-1910318 Draft CR for 38.101-3: EN-DC Pcmax Huawei, HiSilicon R4-1909111 Draft CR for 38.101-3 applicability for intra-band CA and ENDC Huawei, HiSilicon	
2019-12	RAN#86	RP-193032	0074		F	CR for 38.101-3 EN-DC RX Out-of-Band Blocking for shared bands and bands in close proximity	15.8.0
2019-12	RAN#86	RP-193032	0076		F	CR to 38.101-3 Missing Harmonic Mixing MSD for DC_3_n77/n78	15.8.0
2019-12	RAN#86	RP-193032	0078		F	CR for 38.101-3 EN-DC DL Synchronous Carriers	15.8.0
2019-12	RAN#86	RP-193032	0084		F	CR for 38.101-3: Correction to DC Config and dual UL interferer (Rel-15)	15.8.0
2019-12	RAN#86	RP-193032	0086		F	CR for 38.101-3: Correction to EN-DC and NE-DC Configurations (Rel-15)	15.8.0
2019-12	RAN#86	RP-193033	0088	1	F	CR to 38.101-3: clarification of ENDC power class in R15	15.8.0
2019-12	RAN#86	RP-193032	0089	1	F	CR to TS38.101-3: Correction on channel spacing for intra-band EN-DC carriers (section 5.4B.1)	15.8.0
2019-12	RAN#86	RP-193032	0098		F	CR to TS 38.101-3 on inter-band EN-DC configurations including FR2 for five bands (Rel-15)	15.8.0
2019-12	RAN#86	RP-193032	0103		F	CR to TS 38.101-3 on inter-band CA, EN-DC, NE-DC and NR-DC configurations (Rel-15)	15.8.0
2019-12	RAN#86	RP-193032	0104		F	CR to TS 38.101-3: clarification for MPR statement	15.8.0
2019-12	RAN#86	RP-193032	0105		F	CR for TS 38.101-3: Removing MSD requirements for EN-DC combinations due to receiver even order harmonic mixing with UL 3rd harmonic	15.8.0
2019-12	RAN#86	RP-193033	0110		F	CR to TS 38.101-3: adding missing 90MHz channel BW support for n77, n78 related CA	15.8.0
2019-12	RAN#86	RP-193033	0113		F	Removal of brackets from MPR and MSD 38.101-3 REL15	15.8.0
2019-12	RAN#86	RP-193033	0119		F	Pcmax for EN-DC: applicability of NS values and removal of a duty-cycle capability	15.8.0
2019-12	RAN#86	RP-193033	0124		B	CR for TS 38.101-3: Additional out-of-band blocking exceptions for inter-band EN-DC	15.8.0
2019-12	RAN#86	RP-193033	0127		F	CR for 38.101-3: correct MSD exception for DC_2_n78(Rel-15)	15.8.0
2019-12	RAN#86	RP-193033	0135	1	F	CR for 38.101-3: Clarification of the notation for intra-band EN-DC combinations	15.8.0
2019-12	RAN#86	RP-193033	0137		F	CR to 38.101-3-f70 Corrections to MSD Due to Cross-band Isolation for EN-DC in FR1	15.8.0
2019-12	RAN#86	RP-193033	0141		F	CR to 38.101-3 on only synchronous NR-DC between FR1 and FR2 support in Rel-15	15.8.0
2019-12	RAN#86	RP-193033	0142	2	F	CR for 38.101-3 Correction of EN-DC BCS	15.8.0
2019-12	RAN#86	RP-193033	0144		F	CR for 38.101-3 correction for intra-band EN-DC Pcmax	15.8.0
2019-12	RAN#86	RP-193033	0145		F	CR for 38.101-3 intra-band EN-DC MPR/AMPR	15.8.0
2020-03	RAN#87	RP-200396	0163		F	CR for 38.101-3: Correction of MOP tolerance for B41/n41 EN-DC	15.9.0
2020-03	RAN#87	RP-200396	0171		F	CR to TS 38.101-3: corrections on ACS for intra-band contiguous EN-DC	15.9.0
2020-03	RAN#87	RP-200396	0173	1	D	CR to TS 38.101-3: editorial corrections on Rx requirements for intra-band contiguous EN-DC	15.9.0
2020-03	RAN#87	RP-200396	0176	1	F	CR to TS 38.101-3: Correct the intra-band ENDC channel spacing	15.9.0
2020-03	RAN#87	RP-200396	0192		F	CR to TS 38.101-3: editorial correction for output power dynamics for intra-band EN-DC	15.9.0
2020-03	RAN#87	RP-200396	0198		F	CR on correction of 38.101-3 NEDC Ppowerclass (Rel-15)	15.9.0
2020-03	RAN#87	RP-200396	0209	1	F	Rel-15 CR to 38.101-3 for editorial corrections	15.9.0
2020-03	RAN#87	RP-200396	0210		F	CR to 38.101-3 R15 to remove FDM ULSUP combinations	15.9.0
2020-03	RAN#87	RP-200396	0212		F	CR for inter-band ENDC Tx requirement_Rel-15	15.9.0
2020-03	RAN#87	RP-200396	0216	1	F	EN-DC configuration table corrections	15.9.0

2020-06	RAN#88	RP-200985	0239		F	CR for TS 38.101-3: MSD due to UL harmonic	15.10.0
2020-06	RAN#88	RP-200985	0244		F	MOP for interband EN-DC including both FR1 and FR2 REL15	15.10.0
2020-06	RAN#88	RP-200985	0258		F	CR to TS 38.101-3 on configured output power relaxation due to EN-DC (Rel-15)	15.10.0
2020-06	RAN#88	RP-200985	0260		F	CR to TS 38.101-3 on REFSENS relaxation due to EN-DC (Rel-15)	15.10.0
2020-06	RAN#88	RP-200985	0250	1	F	CR for 38.101-3: Corrections for Ppowerclass and referenced sections	15.10.0
2020-06	RAN#88	RP-200985	0266	1	F	CR to TS 38.101-3: Clean up the MSD test point for ENDC(three band)	15.10.0
2020-06	RAN#88	RP-200985	0231	1	F	CR Coexistence cleanup for 38101-3 Rel15	15.10.0
2020-06	RAN#88	RP-200985	0237	1	F	CR for TS 38.101-3: Missing MSD due to cross band isolation	15.10.0
2020-06	RAN#88	RP-200985	0242	1	F	FR1+FR2 CA interband CA BCS support REL15	15.10.0
2020-06	RAN#88	RP-200985	0247	1	B	CR to 38.101-3 MSD due to UL harmonics and intermodulation interference	15.10.0
2020-06	RAN#88	RP-200985	0233	1	F	CR to TS 38.101-3 R15: corrections on ACS for intra-band contiguous EN-DC	15.10.0
2020-06	RAN#88	RP-200985	0235	1	F	CR to TS 38.101-3: editorial corrections on wide band Intermodulation for intra-band contiguous EN-DC in FR1	15.10.0
2020-06	RAN#88	RP-200988	0295		F	CR to remove TBD in 38.101-3	15.10.0
2020-06	RAN#88	RP-200985	0271	2	F	Removal of the Annex modifiedMPR-Behaviour from the NSA specification	15.10.0
2020-06	RAN#88	RP-200985	0239		F	CR for TS 38.101-3: MSD due to UL harmonic	15.10.0
2020-09	RAN#89	RP-201512	0306	1	F	CR for missing IMD MSD in 38.101-3 for DC_1A-41A_n78A, DC_7A-28A_n78A	15.11.0
2020-09	RAN#89	RP-201512	0308		F	Correction to in-band emissions for intra-band contiguous EN-DC	15.11.0
2020-09	RAN#89	RP-201512	0316	2	F	CR to 38.101-3 MSD due to UL harmonics and intermodulation interference	15.11.0
2020-09	RAN#89	RP-201512	0318		F	CR to correct protected band of intra-band EN-DC	15.11.0
2020-09	RAN#89	RP-201512	0322	1	F	CR for TS 38.101-3: FR1 inter-band EN-DC out-of-band blocking UL configuration	15.11.0
2020-09	RAN#89	RP-201512	0325	1	F	Corrections of Japan-related EN-DC co-ex tables for REL-15 combo	15.11.0
2020-12	RAN#90	RP-202485	0369		F	UL output power for spurious response and general Rx	15.12.0
2020-12	RAN#90	RP-202485	0384	1	F	CR to TS 38.101-3: Some corrections on the ENDC	15.12.0
2020-12	RAN#90	RP-202485	0397		F	CR to correct MSD of DC_1A-41A_n77A	15.12.0
2020-12	RAN#90	RP-202485	0399		F	Correction of CR0325 implementation	15.12.0
2020-12	RAN#90	RP-202485	0409	1	F	Correction of applicability of 2Rx requirements	15.12.0
2020-12	RAN#90	RP-202485	0411	1	F	CR 38101-3 R15 Band 10 protection and DC_42_n79 correction	15.12.0
2020-12	RAN#90	RP-202485	0419	1	F	CR for 38.101-3 Correction on EN-DC synchronous carriers (R15)	15.12.0
2020-12	RAN#90	RP-202485	0423	1	F	CR for TS 38.101-3: correction of spurious emission band UE co-existence (R15)	15.12.0
2021-03	RAN#91	RP-210117	0474	1	F	CR for 38.101-3 to introduce a new MSD due to the counter intermodulation interference(Rel-15)	15.13.0
2021-03	RAN#91	RP-210117	0502		F	CR for 38.101-3: clarification of intra-band EN-DC BCS applicability	15.13.0
2021-03	RAN#91	RP-210851	0508	1	F	Correcting FR1-FR2 BCS ambiguity – Interpretation B	15.13.0
2021-06	RAN#92-e	RP-211080	0514	1	F	CR for clarification on interBandContiguousMRDC in TS 38.101-3	15.14.0
2021-06	RAN#92-e	RP-211080	0517	1	F	Corrections to EN-DC spurious emission tables	15.14.0
2021-06	RAN#92-e	RP-211086	0530	-	F	CR to TS 38.101-3[R15]: Addition of UE co-existence requirements for band 40 and n40.	15.14.0
2021-06	RAN#92-e	RP-211080	0534	1	F	Cleanup for UE co-existence 38.101-3 Rel-15	15.14.0
2021-06	RAN#92-e	RP-211088	0572		F	CR to TS38.101-3: Correction on TIB,c description for FR1-FR2 CA	15.14.0
2021-09	RAN#93-e	RP-211921	0430	3	F	CR to 38.101-3 on handling of fallbacks for FR2 CA	15.15.0
2021-09	RAN#93-e	RP-211923	0636		F	Big CR for TS 38.101-3 Maintenance part1 (Rel-15)	15.15.0
2021-12	RAN#94-e	RP-212854	0669		F	Big CR for TS 38.101-3 Maintenance (Rel-15)	15.16.0
2022-03	RAN#95	RP-220337	0700		F	Big CR for TS 38.101-3 Maintenance (Rel-15)	15.17.0
2022-06	RAN#96	RP-221655	0733		F	Big CR for TS 38.101-3 Maintenance (Rel-15)	15.18.0
2022-09	RAN#97	RP-222023	0750		F	Big CR for 38.101-3 maintenance (Rel-15)	15.19.0
2022-10	RAN#97					Editorial change	15.19.1
2022-12	RAN#98-e	RP-223291	0764	1	F	CR for TS 38.101-3 on clarifications for intra-band EN-DC configurations	15.20.0
2022-12	RAN#98-e	RP-223290	0788		F	CR to TS38.101-3[R15] 4Rx MSD for ENDC	15.20.0
2022-12	RAN#98-e	RP-223290	0797		F	R15 CR on inter band ENDC OOBE correction	15.20.0
2022-12	RAN#98-e	RP-223291	0802	1	F	CR on TDD RMC for Intra-band EN-DC - TS 38.101-3	15.20.0
2023-03	RAN#99	RP-230502	0828			CR on Harmonic mixing MSD for DC_8A-n79A (R15)	15.21.0

2023-03	RAN#99	RP-230502	0839		F	CR to R15 TS38.101-3 maintenance for UE co-ex requirements for UL EN-DC	15.21.0
2023-03	RAN#99	RP-230503	0870		F	CR for TS 38.101-3 to introduce DC_20_n28 general description	15.21.0
2023-03	RAN#99	RP-230503	0887		F	CR for TS 38.101-3 on intra-band EN-DC band combination support for DC_(n)41	15.21.0
2023-06	RAN#100	RP-231355	0929		F	UNDONE CHANGES (because of missing changes in Rel-16 CR#0930, it will be implemented in the next version) Rel15 Cat F CR for 38.101-3 Add RI8,8R relaxtion for ENDC with 8 Rx antenas ports for EUTRA bands	15.22.0
2023-06	RAN#100	RP-231355	0936		F	CR to TS38.101-3: Corrections on MOP FR1-FR2 inter-band NR CA and MOP/MPR for ENDC including FR2/FR1 and FR2	15.22.0
2023-06	RAN#100	RP-231356	0963		F	CR to R15 TS38.101-3 for addition of missing MSD requirements for DC_19_n77 and DC_21_n77	15.22.0
2023-06	RAN#100	RP-231356	0969		F	CR for corrections on EN-DC channel bandwidth section	15.22.0
2023-09	RAN#101	RP-232505	0993		F	Rel15 Cat F CR for 38.101-3 Add ? RIB,8R relaxation for EN-DC combos with 8Rx antennas ports for EUTRA bands	15.23.0
2023-09	RAN#101	RP-232501	1004		F	[NR_newRAT-Core] Corrections on the description on the delta Rib value for DCA and DC in REFSEN requirement	15.23.0
2023-09	RAN#101	RP-232487	1013		F	CR for Rel-15 38.101-3 clarification on non-simultaneous Rx/Tx operation for ENDC including CA_n77-n79 and CA_n78-n79	15.23.0
2023-12	RAN#102	RP-233331	1042		F	[NR_newRAT-Core] Correction to 6.5B.3.3.2 (Rel-15)	15.24.0
2024-03	RAN#103	RP-240561	1153		F	(NR_newRAT) Clarifications for FR2 testing with NR-DC and NR-CA	15.25.0
2024-03	RAN#103	RP-240562	1098		F	CR for TS 38.101-3 Rel-15 CAT-F: Introducing missing Rel-15 MSD requirements	15.25.0
2024-03	RAN#103	RP-240563	1128		F	(NR_newRAT-Core) CR to R15 TS 38.101-3 correct PC3 MSD for DC_19A_n77A and DC_3A-19A_n79A	15.25.0
2024-03	RAN#103	RP-240566	1134	1	F	(NR_newRAT-Core) CR for 38.101-3 corrections to EN-DC power class table	15.25.0
2024-06	RAN#104	RP-241382	1225		F	(NR_newRAT-Core) Correct the NOTE for harmonic mixing MSD valid test point	15.26.0
2024-06	RAN#104	RP-241382	1241		F	Clarification of general spurious emission test range in ENDC (R15)	15.26.0
2024-06	RAN#104	RP-241383	1261		F	CR to 38.101-3 on enabling missing UL feature support for inter-band CA between FR1 and FR2	15.26.0
2024-12	RAN#106	RP-243054	1312	1	F	(NR_newRAT-Core) CR for 38.101-3 EN-DC interband 2UL co-ex simplication R15	15.27.0
2025-03	RAN#107	RP-250579	1361	2	F	CR for TS 38.101-3 to correct Test BW configuration of DC_5A-7A_n78A MSD	15.28.0
2025-06	RAN#108	RP-250911	1379	1	F	(NR_newRAT-Core) CR to correct the Band 41 BW used for the DC_5A_41A_n78A MSD in clause 7.3B.2.3.5.2 according to CA_5A-41A - TS38.101-3	15.29.0